Case Study: How can tourism analytics improve revenue and visitor experience?

Introduction

Welcome to this Tourism Data Analysis Case Study. Following the proven APPASA lifecycle — Ask, Prepare, Process, Analyze, Share, and Act — this study walks through the complete journey of transforming raw tourism data into actionable business insights. By the end of this document, you will have a thorough understanding of tourist behaviour patterns, revenue drivers, and strategic marketing opportunities for tourism stakeholders.

Scenario

You are a junior data analyst working on the marketing team of a global tourism analytics platform. The platform collects data on visitors and revenue from tourist destinations around the world. This data helps tourism boards, travel agencies, and destination managers understand travel trends and improve their services.The Chief Strategy Officer believes that analyzing this data can reveal new growth opportunities, such as identifying which destination categories, seasons, and regions attract the most visitors and generate the highest revenue.You have been assigned to analyze a dataset of 5,989 destination records and provide clear, high-level recommendations to support marketing strategies and infrastructure planning.

Characters and Stakeholders

- **Chief Strategy Officer**: Commissioned the analysis to support growth strategy decisions.
- **Marketing Analytics Team**: The team you are part of, responsible for conducting and leading the data analysis.
- **Destination Managers**: End-users who will use the insights to improve and manage their destinations.
- **Government Tourism Boards**: Policy-level stakeholders interested in understanding seasonal and regional tourism trends.

About the Dataset

The tourism dataset captures destination-level records from around the world, including visitor counts, revenue figures, ratings, accommodation availability, and seasonal information. It spans multiple countries and tourism categories, providing a rich foundation for multivariate analysis.

Column	Type	Description
Location	Categorical	Name of the tourist destination
Country	Categorical	Country where the destination is located
Category	Categorical	Type of tourism (Cultural, Adventure, Nature, etc.)
Visitors	Numerical	Number of visitors recorded at the destination
Rating	Numerical	Tourist satisfaction rating on a 1–5 scale
Revenue	Numerical	Revenue generated from tourism activity
Accommodation_Available	Binary	Whether accommodation is available (Yes/No)
Visit_Date	Date	Date of the recorded visit
Season	Categorical	Season of visit (Spring, Summer, Autumn, Winter)

ASK

Every successful data analysis begins with clear questions. In this phase we define the business task, identify the stakeholders, and determine exactly what insights are needed before touching the data.

Business Questions

The following three core questions drive this analysis:

• What are the key trends in global tourist destination usage by category, season, and region?

• How do visitor numbers, ratings, accommodation, and revenue relate to each other?

• How can these insights influence marketing strategy and infrastructure investment for tourism stakeholders?

Case Study Roadmap — Ask

Guiding Questions	Key Tasks
What problem are we trying to solve for tourism stakeholders?	1. Identify and document the primary business task
How can visitor and revenue insights drive business decisions?	2. List all key stakeholders and their interests
Who are the key stakeholders and what do they need?	3. Define the scope of analysis (categories, regions, time period)
	4. Frame the three core analytical questions

Deliverable

A clear statement of the business task with three guiding questions focused on tourism trends, visitor-revenue relationships, and actionable marketing strategy.

Prepare

In this phase the dataset is sourced, evaluated for credibility, and assessed for any limitations before analysis begins.

Guiding Questions

- What data do you need to answer your questions?
- Is the data complete, accurate, and relevant?

Key Tasks

- Collect the tourism dataset (locations, country, category, visitors, rating, revenue, accommodation availability).
- Understand the meaning of each column and its relevance to your business task.

Deliverable

- Description of all data sources used, including what each column represents.

Process

The Process phase covers all cleaning and transformation steps taken to make the data ready for analysis. Every step is documented so that findings can be reviewed and reproduced.

Tool Selection

Python was selected as the primary tool for its flexibility, open-source ecosystem, and ability to handle both data cleaning and visualization within a single Jupyter Notebook environment. Key libraries used:

• Pandas — data loading, cleaning, and manipulation

• Matplotlib / Seaborn — visualization and exploratory plots

• Jupyter Notebook — interactive, reproducible analysis environment

Step	Action	Outcome
1	Loaded dataset with pd.read_csv()	5,989 rows × 9 columns confirmed
2	Checked missing values (df.isnull().sum())	No missing values found in any column
3	Converted Visit_Date to datetime format	Enables time-based and seasonal aggregations
4	Identified duplicate rows (df.duplicated().sum())	Duplicates detected and removed
5	Verified data types via df.info()	Confirmed numerical and categorical types correct
6	Reviewed summary statistics (df.describe())	Confirmed reasonable ranges for Visitors, Revenue, Rating

Case Study Roadmap — Process

Guiding Questions

• What tools are being used and why?

• Have you ensured the data's integrity?

• What steps ensure the data is clean?

• Have you documented the cleaning process?

Key Tasks

1. Check the data for errors and nulls
2. Choose tools (Python / Pandas)
3. Transform data types (datetime conversion)
4. Remove duplicate records
5. Document the entire cleaning process

Deliverable

Fully documented cleaning process: no missing values found, date column converted, duplicates removed, and data types verified — resulting in a clean, analysis-ready dataset.

Analyze

Case Study Roadmap — Analyze

Guiding Questions

• How should data be organized for analysis?

• Has the data been properly formatted?

• What surprises emerged from the data?

• What trends or relationships were found?

• How do insights answer the business questions?

Key Tasks

1. Aggregate data to surface meaningful groups
2. Organize and format data for visualization
3. Perform univariate and multivariate calculations
4. Identify key trends and relationships

Deliverable

Summary of 17 analytical findings covering univariate distributions and 10 multivariate relationships — revealing that visitors, season, category, and accommodation availability are the primary revenue drivers.

With clean data in hand, we perform both univariate and multivariate analyses to uncover patterns, relationships, and anomalies across the nine variables.

The correlation-based bi-variate and multi-variate analysis of the tourism dataset reveals that there is **no significant linear relationship** among the key variables: visitors, ratings, and revenue. All correlation values are extremely close to zero, indicating that these factors operate largely **independently of one another**.

This suggests that an increase in the number of visitors does not necessarily lead to higher revenue, nor does it significantly impact customer ratings. Similarly, higher ratings do not translate into increased revenue, implying that customer satisfaction alone is not a driving force behind financial performance.

Overall, the analysis highlights that tourism outcomes are likely influenced by **external factors** such as pricing strategies, tourist spending behavior, service offerings, and seasonal trends, rather than direct interactions between visitor volume, satisfaction, and revenue.

Share

The Share phase translates analytical findings into clear, audience-appropriate visualizations and narratives that stakeholders can understand and act upon.

Key Visualizations Produced

All visualizations were created using Matplotlib and Seaborn within a Jupyter Notebook environment. The following chart types were used to communicate insights:

Visualization	Insight Communicated
Category Distribution (Count Plot)	Highlights most common and rarest tourism types
Top 7 Countries Bar Chart	Shows geographic concentration of tourism destinations
Visitors Distribution (Histogram +	Reveals skewed visitor distribution with long

KDE)	right tail
Rating Box Plot	Demonstrates generally high satisfaction with few outliers
Revenue Distribution (Histogram + KDE)	Shows revenue concentration in high-earning destinations
Accommodation Availability (Pie Chart)	Quick visual of infrastructure readiness
Season Distribution (Count Plot)	Identifies seasonal demand patterns
Visitors vs Revenue (Scatter)	Positive relationship — more visitors correlates with more revenue
Category vs Revenue (Bar Plot)	Reveals highest-earning tourism categories
Season vs Visitors (Bar Plot)	Peak travel season identification for marketing timing
Rating vs Visitors (Scatter, sample)	Tests whether quality drives popularity
Accommodation vs Visitors (Box Plot)	Quantifies impact of accommodation on visitor counts
Revenue by Country (Horizontal Bar)	Geographic revenue concentration analysis
Visitors by Category (Box Plot)	Variance in visitor volumes across tourism types
Correlation Heatmap	Overall numerical variable relationships at a glance
Rating vs Revenue (Scatter, sample)	Tests quality-revenue link
Season vs Revenue (Bar Plot)	Confirms seasonal revenue alignment with visitor peaks

Guiding Questions

• Were the business questions answered?

• What story does the data tell?

• Who is the audience and how should we communicate?

• Can visualizations support the findings effectively?

Key Tasks

1. Determine the best format for sharing findings
2. Create effective and clear data visualizations
3. Present findings with narrative context
4. Ensure output is accessible to non-technical stakeholders

Deliverable

18 supporting visualizations created in Seaborn/Matplotlib, organized by univariate and multivariate themes, with clear titles and narrative context for stakeholder presentation.

Act

The final phase converts analytical insights into concrete, high-level recommendations that tourism boards, destination managers, and marketing teams can implement.

Guiding Questions

- What actions or recommendations follow from your analysis?
- How can tourism authorities or businesses use these insights?

Key Tasks

- Suggest strategies to promote under-visited destinations.
- Recommend improvements in accommodations or services.
- Propose marketing or digital campaigns to attract more tourists.

Deliverable

Your top high-level insights based on your analysis

Top High-Level Recommendations

Next Steps

1. The organization should introduce premium services, bundled packages, and upselling strategies so that each visitor contributes more revenue, instead of only trying to increase the number of visitors.
2. Instead of mass marketing, the organization should focus on attracting tourists who are more likely to spend, such as luxury travelers or niche segments, using personalized marketing campaigns.
3. The organization should link good ratings with loyalty programs, repeat visit incentives, and paid exclusive experiences to turn customer satisfaction into actual financial gain.
4. The organization should expand beyond basic tourism income by adding events, food services, and partnerships, ensuring multiple sources of revenue.
5. The organization should adopt season-based and demand-based pricing, increasing prices during peak seasons and offering discounts or packages during off-seasons to maximize overall revenue.

Wrap up

Congratulations on completing the Tourism Data Analysis Case Study! If you like, complete one of the other case studies to continue growing your portfolio. Or, use the steps from the ask, prepare, process, analyze, share, and act Case Study Roadmap to create a new project all your own. Best of luck on your job search!